Developing a Tiny Transformer Language Model

1. Load the dataset

   Load the roneneldan/TinyStories dataset from Hugging Face.

   Explore the dataset. Display several stories and examine their text length and structure.

2. Preprocess the text

   Select a small subset of stories and tokenize the text at the character level or use a tokenizer.

3. Preparing training data

   Create input sequences and corresponding target sequences for next-token prediction.

4. Build the Transformer architecture

   Define token embeddings, positional embeddings, masked multi-head self-attention, feed-forward layers, residual connections, and layer normalization.

5. Initialize the model

   Initialize the Transformer parameters and configure the loss function and optimizer.

6. Train the model

   Feed the input sequences into the Transformer, predict the next tokens, calculate cross-entropy loss, perform backpropagation, and update the model parameters.

7. Monitor training

   Record training and validation loss across epochs and plot the loss curves.

8. Evaluate the model

   Calculate validation loss and perplexity to assess how well the model predicts the next token.

9. Generate stories

   Provide a starting prompt, such as “Once upon a time”, and generate text using the trained Transformer.

10. Experience with model settings

    Change the number of Transformer layers, attention heads, or training epochs and observe the effect on loss and generated text.

11. Save the trained model

    Save the model parameters and configuration for future text generation.

Optional: - Build a chat window. Use gradio

Discuss what mechanisms are used to improve performance.